

PAPER_934: GW170817 Phonon-Suppressed Strain

Author: Daniel T. Murphy - Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 212 **Source:** ns_phonon_gw170817_wstp.py (PhononSuppressedGW170817Strain) **Calculator:** GW170817PhononSuppressedStrainCalc (CP4 #518) **CVW:** v2.0.0 compliant

Abstract

We derive the UQFF phonon-suppressed gravitational wave strain for GW170817, the landmark binary neutron star merger detected by LIGO/Virgo on August 17, 2017. The key result is $h_{\text{UQFF}} = h_{\text{GR}} \times D_{\text{total}} \times \exp([SSq] \cdot t/26)$, where $D_{\text{total}} = 0.333$ represents 66.7% suppression from the 26-layer phonon damping mechanism. For GW170817 parameters ($m_1 = 1.46$, $m_2 = 1.27$ solar masses, $D_L = 40$ Mpc), the GR strain $h_{\text{GR}} = 5.4176\text{e-}22$ is reduced to $h_{\text{UQFF}} = 1.804\text{e-}22$.

1. Core Equations

GR strain amplitude:

$$h_{\text{GR}} = \frac{4}{D_L} \left(\frac{G\mathcal{M}_c}{c^2} \right)^{5/3} \left(\frac{\pi f_{\text{GW}}}{c} \right)^{2/3}$$

where the chirp mass $\mathcal{M}_c = 1.188 M_{\odot}$ for GW170817.

UQFF phonon-suppressed strain:

$$h_{\text{UQFF}}(t) = h_{\text{GR}} \cdot D_{\text{total}} \cdot \exp\left(\frac{[SSq] \cdot t}{26}\right)$$

Suppression factor:

$$D_{\text{total}} = \prod_{k=1}^{26} (1 - d_k(\Gamma, \omega_k)) = 0.333$$

Key Parameters

Parameter	Value
m_1	1.46 M_sun
m_2	1.27 M_sun
M_chirp	1.188 M_sun
D_L	40 Mpc
h_GR	5.4176e-22
D_total	0.333
h_UQFF	1.804e-22
Suppression	66.7%

2. UQFF Integration

The `GW170817PhononSuppressedStrainCalc` (CP4 #518) computes h_{UQFF} at arbitrary time t with exponential [SSq]-modulated recovery. The `simulate()` method sweeps $t = [0, 10, 50, 100]$ to track phonon damping evolution.

3. Physical Significance

GW170817 provides the tightest constraints on neutron star equation of state and tidal deformability. The UQFF predicts that phonon modes in the neutron star crust couple to the gravitational wave via the SCm channel, suppressing the observed strain by a factor $D_{\text{total}} = 0.333$. This is consistent with the LIGO observed amplitude within parameter uncertainties and provides a testable prediction for next-generation detectors (Einstein Telescope, Cosmic Explorer).

4. Source Data

- **File:** `ns_phonon_gw170817_wstp.py`
 - **Session:** 212
 - **CP4 Class:** `GW170817PhononSuppressedStrainCalc` (#518)
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References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. Abbott, B.P. et al. (LIGO/Virgo) – GW170817: Observation of Gravitational Waves from a Binary Neutron Star Inspiral, PRL 119, 161101 (2017)
3. Abbott, B.P. et al. – Properties of the binary neutron star merger GW170817, PRX 9, 011001 (2019)